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Roll No

IT-831 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Real Time Systems
(Elective - III)
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain task - scheduling algorithm, what is the difference between RM and EDF Algorithm.
OR
Explain FAB Algorithm in detail.
2. Explain the factors for estimating program run times. Calculate total execution time of the following code.
L1: a:=b*c;
L2: b:=d + e;
L3: d:=a - f;
3. Define the following with respect to real time programming languages with example
 - a) Global and Local variables:
 - b) Modularity
 - c) Exception handling
4. Explain Next-Fit algorithm for RM scheduling.
5. What is Static Pairing? Explain with example.
6. Why run-time is important in real time systems? On which factors, execution time of any program depends?
OR
What are Malicious or Byzantine faults/failures? Discuss the different redundancy issues involved in fault tolerant RTS.
7.
 - a) What are Pre-emptive and Non-Preemptive scheduling? Explain with suitable example.
 - b) Describe myopic offline scheduling algorithm.
8. Write short notes on any two
 - i) Bidding algorithm
 - ii) Low level programming
 - iii) Polled bus
